6 Similarity indices and matching representations

We now turn our attention to the relation of similarity and matchability of representations. Finding an appropriate similarity index for neural representations is an active area of research [Gretton et al., 2005, Raghu et al., 2017, Morcos et al., 2018, Kornblith et al., 2019]. In the pursuit of a suitable notion, one seeks to capture the right invariance and equivariance properties to provide a tool with theoretical guarantees and practical applicability. Our brief summarizing statement in this section acts as a warning for the end-users of statistical similarity indices:

By their nature, similarity indices depend on the task only indirectly, and their values are not necessarily indicative of task performance on a stitched network.

6.1 Similarity indices

Given two sets of activations $A = (a_i)_{i=1}^n, B = (b_j)_{j=1}^n$, in the activation spaces $a_i \in \mathcal{A}_1 = \mathbb{R}^s, b_j \in \mathcal{A}_2 = \mathbb{R}^r$, a *similarity index* $s(A, B) \in [0, 1]$, measures the similarity of the two data representations, i.e. it is a statistic on the observation space $s : \mathcal{A}_1 \times \mathcal{A}_2 \to [0, 1]$. We now briefly review the similarity indices under consideration. First, the least squares matching M_{LS} defined earlier can be used to define a statistic that can be used as a similarity index [Kornblith et al., 2019]:

Definition 6.1. The coefficient of determination *of the best fit is given in terms of the optimal least squares matching $M_{LS} = A^\dagger B$ as:*

$$R^2_{LR}(A, B) = 1 - \frac{\|AM_{LS} - B\|_F^2}{\|B\|_F^2}.$$

Let k, l be positive definite kernels on a space $\mathcal{X}$ and let $K_{ij} = k(x_i, x_j)$ and $L_{ij} = l(x_i, x_j)$ be the Gram matrices of a finite sample $x_i \in \mathcal{X}$, $i = 1, \dots, m$ drawn from a distribution on $\mathcal{X}$. The (empirical) *alignment of the Gram matrices* K, L was defined by Cristianini et al. [2001] as

$$\mathrm{KA}(K, L) = \frac{\langle K, L\rangle_F}{\|K\|_F \cdot \|L\|_F}.$$

The importance of centering the kernels in practical applications was highlighted in Cortes et al. [2012]. Given a Gram matrix K, denote the *centered Gram matrix* by $K^c := CKC$, where $C = I_m - \frac{\mathbf{1}\cdot\mathbf{1}^T}{m}$ is the centering matrix of size m. The *Centered Kernel Alignment of the Gram matrices* K, L is defined [Cortes et al., 2012] as $\mathrm{CKA}(K, L) = \mathrm{KA}(K^c, L^c)$. If the kernels are linear on their respective feature spaces, and if X, Y are samples of the distribution in these feature spaces, then $\mathrm{CKA}(K_X, L_Y) = \mathrm{KA}(K_{XC}, L_{YC})$, using that $K_X^c = K_{XC}$. In particular, in the case of linear kernels CKA is also referred to as the *RV coefficient* [Robert and Escoufier, 1976]. Furthermore, if X, Y are the *centered* matrices representing the samples of the distribution in the feature space, CKA can be given by the following definition:

Definition 6.2. *The* empirical (linear) Centered Kernel Alignment *or* RV coefficient *of the samples X, Y centered in feature space is*

$$\mathrm{RV}(X, Y) := \mathrm{CKA}(K_X, K_Y) = \frac{\|X^T Y\|_F^2}{\|X^T X\|_F \cdot \|Y^T Y\|_F}.$$

In our context, the probability distribution describes the data distribution in $\mathcal{X}$, the feature spaces are the activation spaces and the feature maps are the representation maps $R : \mathcal{X} \to \mathcal{A}$.

As a measure of similarity of representations, CKA was introduced and used successfully in Kornblith et al. [2019]. They also considered other similarity indices such as CCA, SVCCA, PWCCA [Raghu et al., 2017, Morcos et al., 2018] in their experiments, but they found that the other similarity indices do not succeed in identifying similar layers of identical networks trained from different initializations, whereas CKA manages to do so. For this reason, we chose to focus our attention on CKA.

6.2 Experiment 3: Similarity indices and task performance

In this set of experiments, we monitor the CKA, $\mathrm{R}^2_{\mathrm{LR}}$, CCA and SVCCA indices throughout the training process of the stitching layer trained with task loss matching. We measure these values

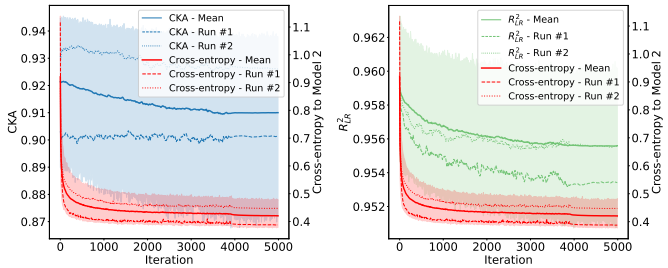

Figure 4: CKA, $\mathrm{R}^2_{\mathrm{LR}}$ and cross-entropy values over the training iterations of task loss matching started from the optimal least squares matching. Tiny-10 network, stitching at Layer 3. Bold lines are averages of 10 runs, with bands representing standard deviations; dotted and dashed lines correspond to two specific runs to show the individual characteristics of the training of the stitching layer. Further experiment details are in Appendix A.5.

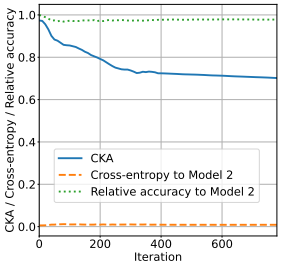

Figure 5: CKA and performance measures throughout the training for a Frankenstein task with a loss that explicitly penalizes CKA value while maintaining the model performance. Stitching is at Layer 2.0 of a 3-wide ResNet.

between Model 2 activations and the transformed activations of Model 1. Figure 4 shows the results for CKA and $\mathrm{R}^2_{\mathrm{LR}}$. We observe that CKA has a large variance from the first step of the training process, and it is decreasing while the performance of the stitched model is increasing.

This experiment also highlights that $\mathrm{R}^2_{\mathrm{LR}}$ can correlate inversely with the performance: as the training proceeds with decreasing cross-entropy, the $\mathrm{R}^2_{\mathrm{LR}}$ values also decrease. Indeed, since we initialized the task loss training from the optimal least squares matching, the distances naturally must increase under any training process. However, this highlights the fact that the least squares matching is ignorant of the decision boundaries in a neural network.

CCA is invariant to invertible linear transformations, hence it ignores our linear stitching transformation during the training. SVCCA also performs CCA on truncated singular value decompositions of its inputs. Because its value only depends on the SVD step, it performs quite big jumps during the training, but on average it does not correlate with the task performance. See further results and experimental details in Appendix A.5.

6.3 Experiment 4: Low CKA value with high accuracy

We now show that it is easy to align representations in a way that results in high task performance while the CKA value at the alignment is small. The setup is the following: we take one trained model and stitch it with *itself*, utilizing a linear stitching layer initialised with the identity mapping. With this initialization we essentially reproduce the original model, thus, this trivially results in a Frankenstein network with the same performance as the original model. We train this stitching layer with an additional loss term that penalizes the CKA value at the stitching, meanwhile we maintain the task performance with the usual cross-entropy term. We use the CKA value itself as the penalty, as it is differentiable by definition and thus suitable for gradient-based optimization.

Figure 5 shows the outcome of such a training process. We find that the CKA value drops by a large amount (achieving lower than 0.7), whereas the accuracy remains essentially unchanged.

As a baseline, we provide the CKA values between representations of the transformed activations of Model 1 and Model 2 averaged over 10 different network pairs and all layers. ResNet: 0.922 ± 0.098, Tiny-10: 0.883 ± 0.050. For further baseline CKA values we also refer to Kornblith et al. [2019].

6.4 Discussion

Notions of representational similarity and functional similarity have different motivations, and thus, aim to capture different invariances. Part of these invariances could be compatible or incompatible. For example, assuming that a similarity index s is invariant under a given class $\mathcal{C}$ of transformations, it is not surprising that the high value of a similarity index $s(A, B)$ does not indicate the relative performance of the activations A and B: indeed, given a set of activations A, $s(AF, A) = 1$ for any $F \in \mathcal{C}$, while the task loss of the transformed activations AF typically varies greatly. On the other hand, it is somewhat surprising that high-performance matchings can have a significant drop in the values of meaningful similarity indices such as CKA.

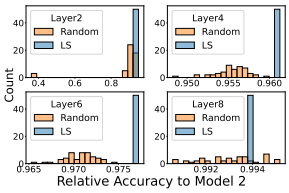

Figure 6: Relative accuracies by initialization type.

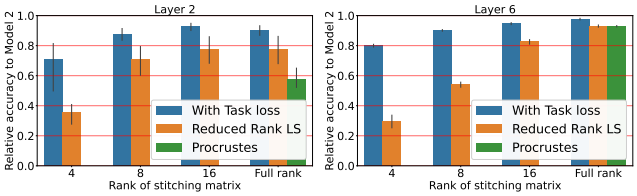

Figure 7: The performance of the low-rank analogues of least squares and task loss matchings in terms of relative accuracy.

The inadequateness of CKA as a direct matching method CKA is not directly suitable as an objective for direct matching, since it is invariant to orthogonal transformations. A stitching with an optimal CKA value can be arbitrarily transformed via orthogonal transformations without affecting CKA, but most of these equivalent solutions are unsuitable for the matching task as trained models are not invariant for arbitrary rotations in the activation spaces.

7 Low rank representations — a gap between SVD and task loss matching

In this experiment, we compare the effectiveness of the low-rank analogues of the above presented least squares matching and task loss matching. More precisely, we apply SVD directly in the activation space to obtain the low-rank least squares matching, and use a *bottleneck* stitching layer to realise a low-rank matching with the task loss.

Figure 7 depicts the results for two selected layers of Tiny-10 with prescribed ranks 4, 8, and 16. (We observed similar results for the other layers, and also for ResNet-20.) It is clear that task loss matching outperforms least squares matching in the activation space when the rank of the transformation is fixed. This suggests that the objective for the low-rank approximation of explaining most of the variance in the activation space (i.e., the objective of SVD) is not satisfactory to attain the highest possible performance with a prescribed rank. In other words, the magnitudes of the principal components are not in direct correspondence with the information content of the latent directions. Ultimately, this suggests that similarity indices that use SVD directly in the activation space — either as a preprocessing step or as another key element in the computation — might fail to capture some desired aspects of the representation. Details and figures for the other layers are in Appendix A.

8 Structure of stitching layers: uniqueness, mode connectivity, sparsity

The previous experiments show that the stitching layer can be trained to achieve the same performance as the original Model 2. What is the structure of the obtained stitching matrix? In this chapter, we explore the question of sensitivity with respect to initialization, uniqueness properties of the stitching layer and the relationship of direct and task loss matching with sparsity.

Experiment 5: Dependence of accuracy on initializations Are stitching layers obtained from different initializations the same? A crude measure of this is the performance of the stitching layer on the stitched network. In this experiment, we fix two identical network architectures trained from different initializations. We train the stitching layer on task loss from 50 random and 50 optimal least squares initializations, and compare their performance. We found that in most cases the stitched networks achieve the same order of accuracy, although not always; the training process might get stuck at poor local minima. However, the optimal least squares matching initialization performs consistently well. Figure 6 shows a histogram of the results. For further details, see Appendix A.

Experiment 6: Linear mode connectivity of transformation matrices We also investigated the uniqueness properties of the stitching layer by examining the linear mode connectivity of the found transformations. In particular, on a given layer of two trained copies of Tiny-10, we trained stitching transformations from different random initializations. We only kept the transformations achieving a relative accuracy of at least 80% — there is no reason to expect mode connectivity properties where there is a large difference in relative accuracy. We then evaluated the relative accuracy of the stitched network with the transformation matrix $M_\lambda = \lambda M_1 + (1-\lambda) M_2$ for pairs of transformation matrices M_1 and M_2, and different values of $\lambda \in [0, 1]$. The results showed that mode connectivity